



Kvaerner Oil & Gas

Project : Huldra EPC Wellhead Platform Topsides

Title : System Control Diagram, HVAC sequences

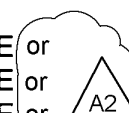
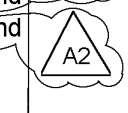
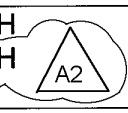
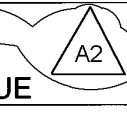
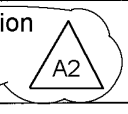
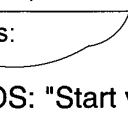
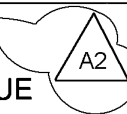
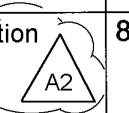
Doc. No : C025-W-HO77-J-*E-003-01

KOG Project No : C131

Statoil Project No : C025

A2	11.01.10	AS BUILT ACC. TO AO. 20958634	KRPAS	EDARN	STTEI	VH
A1	30.09.01	AS BUILT	EiL	V/B.	JD	
0	20.03.01	ISSUED FOR CONSTRUCTION	Buk			
05D	01.08.00	ISSUED FOR IMPL OF MODPACK 2	SNo			
04D	05.06.00	ISSUED FOR FAT	EiL			
02D	26.11.99	ISSUED FOR DESIGN	SNo			
Revision	Date	Revision Description	Prepared	Checked	Approved	Comp. Appr.

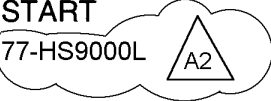
START HVAC SEQUENCE

Phase	Step	Actions	Outputs to	Next step	Wait for (before entering the next step)	Jump condition	Jump to step	Remarks
Start permissives 77-HS9000H 	0	NA	NA	1 OR 2	System not available: 77-GK 50A.XE or 77-GK 50B.XE or 77-GK 52A.XE or 77-GK 52B.XE = FALSE 	System available: 77-GK 50A.XE and 77-GK 50B.XE and 77-GK 52A.XE and 77-GK 52B.XE = TRUE	2	Manual start from OS, OR Automatic start, ref. SCD 77-001. See also NOTE 2. Lack of AVAILABLE-signal from fans, will inhibit start of HVAC.
Abort because equipment not available	1	Warning on OS	77-US 9001.X	END		NA	NA	
Set system to auto	2	1-6: Set system in auto	1: 77-GJ 51.LA (via Z173) 2: 77-GJ 52.LA (via Z174) 3: 77-GK 50A.LA 4: 77-GK 50B.LA 5: 77-GK 52A.LA 6: 77-GK 52B.LA	3	NA	NA	NA	
Open all dampers Indication on OS: "Åpne spjeld"	3	1: Open all dampers 2: Start timer C992– dampers timeout – 5 sec	1: 77-GS 50.XH (F03), 77-GS 51.XH (F03), 77-GS 52.XH (F03), 77-GS 60.XH (F03), 77-GS 61.XH (F03), 77-GS110.XH (F03) 77-GS111.XH (F03)  (One signal sent to F03 via C988) 2: C992.X	4 OR 5	Timer expired: C992.Y = TRUE	All dampers confirmed open: 77-GS 50.BCH(F03) and 77-GS 51.BCH(F03) and 77-GS 52.BCH(F03) and 77-GS110.BCH(F03) and 77-GS 60.BCH(F03) and 77-GS111.BCH (F03) and 77-GS 61.BCH(F03) =TRUE  (One signal received from F03 via M533.1, "F03_HVAC_dampers open") 	5	
Abort because of damper malfunction	4	1-2: Start HVAC Stop sequence 3: Warning on OS	1: 77-HS 9000L.LA (via Z341) 2: 77-HS 9000L.XH 3: 77-US 9002.X	END	NA	NA	NA	
Start supply fans:  Indication on OS: "Start vifter i luftinntak"	5	1-2: Start fans on A side 3: Start timerC993- fans timeout – 10 sec	1: 77-GK 50A.XP1H  2: 77-GK 50B.XP1H 3: C993.X	6 OR 7	Timer expired: C993.Y = TRUE	Supply fans running:  77-GK 50A.BCH and 77-GK 50B.BCH = TRUE	7	
Abort because supply fans malfunction 	6	1-2: Start HVAC Stop sequence 3: Warning on OS	1: 77-HS 9000L.LA 2: 77-HS 9000L.XH 3: 77-US 9003.X	END			NA	
								
Start extract fans:  Indication on OS: "Start vifter i luftavtrekk"	7	1-2: Start fans B side	1: 77-GK 52A.XP1H 2: 77-GK 52B.XP1H	8 OR 9	Timer expired: C993.Y = TRUE	Extract fans running:  77-GK 52A.BCH and 77-GK 52B.BCH = TRUE	9	
Abort because of extract fans malfunction 	8	1-2: Start HVAC Stop sequence 3: Warning on OS	1: 77-HS 9000L.LA 2: 77-HS 9000L.XH 3: 77-US 9004.X	END		NA	NA	
Ten seconds time delay Indication on OS: "Pause i 10 sekund"	9	1: Start timer C991 – 10 sec	1: C991.X	10	Timer expired: C991.Y = TRUE	NA	NA	

Switch on heaters Indication on OS: "Slå på varme"	10	1-2: Start heaters 3: Start timer C994- heaters timeout – 5 sec 4: Set the cooling coil temp. contr. to auto	1: 77-GJ 51.XP1H (via C986) 2: 77-GJ 52.XP1H (via C984) 3: C994.X 4: 77-TIC 2004.LA	11 OR END	Timer expired: C994.Y = TRUE	Heaters running: 77-GJ 51.BCH and 77-GJ 52.BCH = TRUE	END	
Alarm because of heaters malfunction	11	Warning on OS	77-US 9005.X	END		NA	NA	
END	END	NA	NA	NA		NA	NA	

Note 1: The sequence is to be implemented in the PCS-node. Fire dampers are handled by the F&G-node.
Note 2: Following loss of power supply by failure/autochangeover of a generator (82-ES2001A/B), the START HVAC SEQUENCE shall be *automatically* initiated. Output from pulsblock C618 active when power comes back.

STOP HVAC SEQUENCE

Phase	Step	Actions	Outputs to	Next step	Wait for	Jump condition	Jump to step	Remarks
START 	0	Set all fans and heaters to auto	1: 77-GJ 51.LA (via Z173) 2: 77-GJ 52.LA (via Z174) 3: 77-GK 50A.LA 4: 77-GK 50B.LA 5: 77-GK 52A.LA 6: 77-GK 52B.LA	1	NA	NA	NA	Manual start from OS, OR Automatic start, ref. SCD 77-001
Switch off heaters Shown on OS: "Slå av varme"	1	1-2: Stop command to heaters 3: Cooling oil temp. contr. to manual	1: 77-GJ 52.XP1L 2: 77-GJ 51.XP1L 3: 77-TIC 2004.LM	2	Heaters confirmed off: 77-GJ 52.BCL and 77-GJ 51.BCL = TRUE	NA	NA	
1 minute time delay On OS: "Pause i 10 s"	2	Start timer Z425	Z425:X	3	Timer expired: Z425=TRUE	NA	NA	
Stop supply and extract fans Shown on OS: "Stopp vifter"	3	1-4: Stop fans	1: 77-GK 50A.XP1L 2: 77-GK 50B.XP1L 3: 77-GK 52A.XP1L 4: 77-GK 52B.XP1L	END	Fans confirmed stopped: 77-GK50A.BCL and 77-GK50B.BCL and 77-GK52A.BCL and 77-GK52B.BCL	NA	NA	
END	END	NA	NA	NA		NA	NA	

Note 1: The sequence is to be implemented in the PCS-node. Fire dampers are handled by the F&G-node.